

Exercise 1: Lenz's rule

The induced voltage always gives rise to a current whose magnetic field opposes the original change in magnetic flux.

- a) The magnetic field of the pole magnet points towards the right through the coil, and through its motion the flux in the coil will be reduced. The induced current therefore has to produce a field that also points towards the right and therefore has to flow from left to right through the resistor. The effect is that the magnet will experience a force pulling him back towards the coil, slowing his motion.
- b) The field building up after connecting the switch will point towards the left. The field induced by the loop to the right therefore has to point opposite to it, to the right, and therefore the induced current flows from back to front through R .
- c) The magnetic field of the conductor points into the plane of the figure at the location of the loop. If the current in the conductor is reduced, also the flux through the loop will be reduced. The induced current therefore has to build up a field opposing this change, i.e. pointing in the same direction as the original one and therefore the current has to flow in clockwise direction and thus from left to right in the resistor.
- d) According to the right-hand rule and the Lorentz force the magnetic field has to point into the plane of the figure.

Exercise 2: Conductor loop at High Voltage line

- a) The magnetic field induced by a current in a straight conductor has a radial symmetry and can be written as follows:

$$B = \mu_0 \frac{I}{2\pi r} \quad (1)$$

To describe the magnetic field at the two desired distances, we replace r with the respective distances:

$$B_d = \mu_0 \frac{I}{2\pi d} = 0.048 \text{ T} \quad (2)$$

$$B_{d+b} = \mu_0 \frac{I}{2\pi(d+b)} = 0.003 \text{ T} \quad (3)$$

$$(4)$$

- b) To calculate the flux we use the following term:

$$\Phi = \int_A \vec{B} d\vec{\alpha}, \quad (5)$$

where A is the area enclosed by the loop and α is a piece of the area. B and A stand perpendicular to each other, therefore for the calculation we can use the absolute values

$$\Phi = \int_A \vec{B} d\vec{\alpha} = \int_0^a dy \int_d^{d+b} B(x) dx \quad (6)$$

$$\Phi = a\mu_0 \frac{I}{2\pi} [\ln(x)]_d^{d+b} = a\mu_0 \frac{I}{2\pi} \ln\left(\frac{d+b}{d}\right) \quad (7)$$

Here, we used the fact that the origin of the y -axis can be freely chosen and that B does not depend on y .

- c) To calculate the induced current in the conductor loop, when the current of the high voltage line is reduced, we calculate first the electromotive force:

$$\mathcal{E} = -\frac{d\Phi}{dt} \quad (8)$$

Since only the magnetic field changes but not the enclosed area, this is an induction of second type. The time dependence of the magnetic flux is contained in the current I . This means

$$\Phi(t) = a\mu_0 \frac{I(t)}{2\pi} \ln\left(\frac{d+b}{d}\right) \quad (9)$$

with

$$I(t) = I_0 \left(1 - \frac{t}{\Delta t}\right) \quad (10)$$

$$I(t) = I_0 \left(1 - \frac{t}{4 \cdot 10^{-4} \text{ s}}\right) \quad (11)$$

$$\frac{dI}{dt} = -\frac{I_0}{4 \cdot 10^{-4} \text{ s}} \quad (12)$$

Therefore we get

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{d(a\mu_0 \frac{1}{2\pi} \ln(\frac{d+b}{d}) I(t))}{dt} = -a\mu_0 \frac{1}{2\pi} \ln\left(\frac{d+b}{d}\right) \frac{dI(t)}{dt} \quad (13)$$

$$\mathcal{E} = -\frac{a\mu_0}{2\pi} \ln\left(\frac{d+b}{d}\right) \left(-\frac{I_0}{\Delta t}\right) = \frac{a\mu_0 I_0}{2\pi \cdot 4 \cdot 10^{-4} \text{ s}} \ln\left(\frac{d+b}{d}\right) \quad (14)$$

To calculate the induced current out of this, we apply Ohm's law with the electromotive force representing the voltage V :

$$\frac{\mathcal{E}}{R} = I = \frac{a\mu_0 I_0}{2\pi \Delta t \cdot R} \ln\left(\frac{d+b}{d}\right) \quad (15)$$

Inserting all the variables, we get an induced current of

$$\frac{0.2 \text{ m} \mu_0 2400 \text{ A}}{8\pi \cdot 10^{-4} \text{ s} \cdot 1.2 \cdot 10^{-4} \Omega} \cdot \ln(16) \approx 5540 \text{ A}. \quad (16)$$

- d) The force acting on the conductor loop is the Lorentz force. Among others, it is built of the cross product between the direction of the current and the magnetic field. Therefore, the right-hand rule can be used to determine in which direction the force acts for the individual sections: the upper and lower conductor sections experience a force upwards and downwards respectively. As the magnetic field along the high-voltage line is constant, we already know that they will

experience the same force, but in opposite directions due to the reversed direction of the current. The two forces will therefore cancel each other out.

Accordingly, the forces for the left and right conductors act to the left and to the right. However, it should be noted here that the left-hand conductor feels a stronger force due to the stronger magnetic field closer to the high-voltage line.

To calculate the force, we can use the formula for the Lorentz force:

$$\vec{F}_L = q(\vec{v} \times \vec{B}) \quad (17)$$

$$I = \frac{dq}{dt} \rightarrow q = I \cdot t \quad (18)$$

$$\vec{v} = \frac{d\vec{l}}{dt} \rightarrow \vec{v} = \frac{\vec{l}}{t} \quad (19)$$

$$\vec{F}_L = I(\vec{l} \times \vec{B}) \quad (20)$$

As the forces for the upper and lower conductors balance each other out, we only consider the left and right conductors. The line segments of these conductors are always perpendicular to the magnetic field, which is why the cross product can be replaced by the product of the magnitudes. We already know the two magnetic fields from before although it should be noted that the current of the high-voltage line is now time-dependent.

$$B_d(t) = \mu_0 \frac{I(t)}{2\pi d} = 0.048 \text{ T} \quad (21)$$

$$B_{d+b}(t) = \mu_0 \frac{I(t)}{2\pi(d+b)} = 0.003 \text{ T} \quad (22)$$

$$(23)$$

The numeric values refer to the time $t = 0$. The length of the conductor is also given by a . We therefore get the following result with I_S and I_H the currents in the loop and in the high-voltage line:

$$(\vec{F}_{Ll} - \vec{F}_{Lr}) = I_S \cdot a \cdot (B_l(t) - B_r(t)) \quad (24)$$

$$= \frac{I_S \cdot a \cdot \mu_0 \cdot I_H(t)}{2\pi} \cdot \left(\frac{1}{d} - \frac{1}{d+b} \right) \quad (25)$$

$$(26)$$

By inserting the values, we get

$$\approx \frac{5540 \text{ A} \cdot 20 \text{ cm} \mu_0}{2\pi} \cdot \left(\frac{15}{16} \right) \cdot 2400 \text{ A} \left(1 - \frac{t}{\Delta t} \right) \quad (27)$$

$$= 49.830 \cdot \left(1 - \frac{t}{4 \cdot 10^{-4} \text{ s}} \right) \text{ N} \quad (28)$$

In this calculation, only the magnitude was used. The direction of the force, as noted before is towards the high power line.

Exercise 3: Semi-circle generator (14.5 Points)

- a) The magnetic flux $\phi(t)$ through the loop can be expressed as

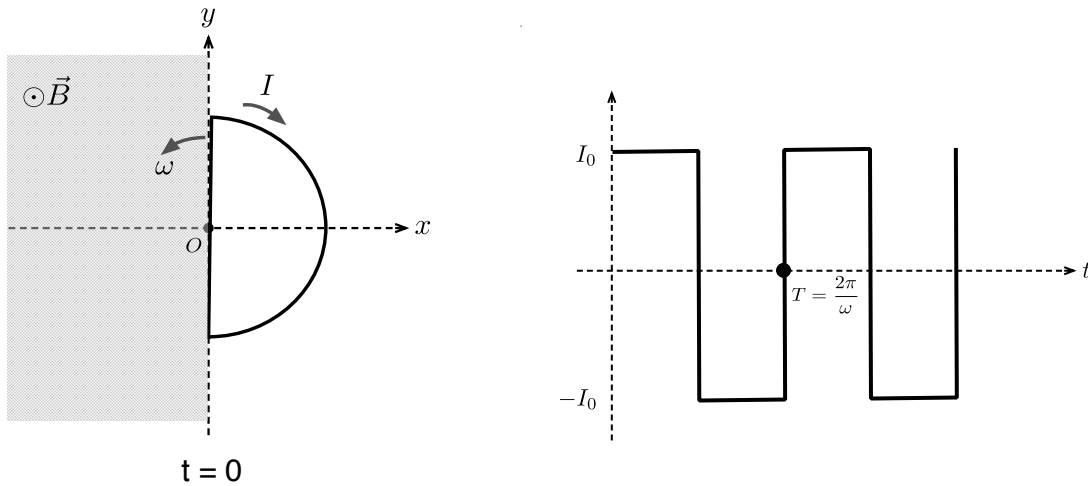
$$\phi(t) = B \cdot A(t) = B \frac{\pi a^2}{2} \frac{\theta(t)}{\pi} = B \frac{\pi a^2}{2} \frac{\omega t}{\pi}, \quad (29)$$

where $A(t)$ is the area of the loop that superposes with the magnetic field. This area can be expressed as the area of the loop $\frac{\pi a^2}{2}$ scaled with the angle $\theta(t) = \omega t$.

- b) For times $t \in [0; T/2]$ the flux through the loop is increasing as the area with non-vanishing magnetic field in the loop is increasing. As a consequence of the Lenz rule, the induced current has to produce an induced magnetic field that decreases the external B -field. Hence a current will be induced clockwise in the loop that produces a magnetic field in $-\hat{z}$ (into the plane) as shown in the figure below (left).

For times $t \in [T/2; T]$ the flux through the loop is decreasing. In this time window a current will be induced counterclockwise in the loop that increases the external field. The induced current produces a magnetic field in $+\hat{z}$ (out of the plane).

Rotation of the loop thus leads to alternating current as shown in the figure below (right) where we defined the current flowing in clockwise direction as positive.



- c) First we need to calculate the induced electromotive force (we neglect here the sign because it was implicitly discussed already above):

$$|\epsilon(t)| = \frac{d\phi(t)}{dt}, \quad (30)$$

$$= \frac{d\left(B \frac{\pi a^2}{2} \frac{\omega t}{\pi}\right)}{dt}, \quad (31)$$

$$= B \frac{\pi a^2}{2} \frac{\omega}{\pi} \frac{dt}{dt} \quad (32)$$

$$= B \frac{a^2}{2} \omega. \quad (33)$$

Taking into account the resistance R of the loop the current in the loop becomes

$$I_0 = \frac{|\epsilon|}{R} = \frac{Ba^2}{2R}\omega. \quad (34)$$

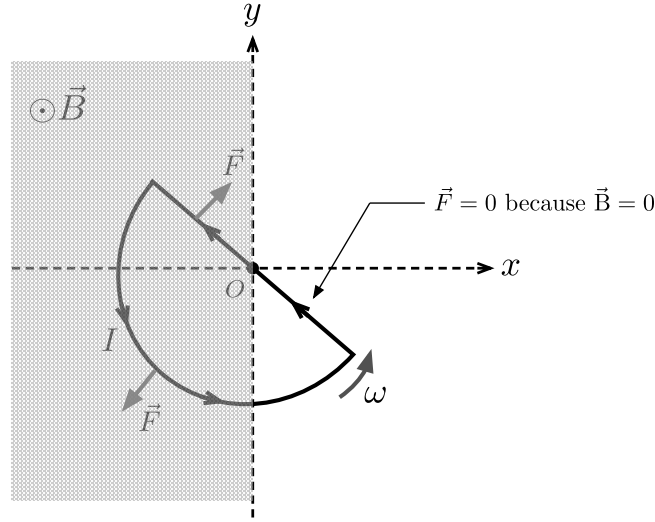
- d) The external total torque results from the sum of the torque exerted on the infinitesimal parts forming the loop,

$$d\vec{M} = \vec{r} \times d\vec{F}, \quad (35)$$

where $d\vec{F}$ is the force exerted on the infinitesimal part of wire $d\vec{l}$ given by

$$d\vec{F} = Id\vec{l} \times \vec{B}. \quad (36)$$

The loop can be decomposed in the two parts: the straight segment and the semi-circle



On the straight segment:

- Half of the segment (the lower part in the above figure) is having no force because there $B = 0$
- The other half is having a torque in the $+\hat{z}$ direction

On the semi-circle, the force $\vec{F}$ is in the direction of $\vec{r}$ so that it does not contribute to the torque.

The total torque is thus given by half of the straight segment:

$$\vec{M} = \int_0^a d\vec{M} = \int_0^a \vec{r} \times d\vec{F} = \int_0^a \vec{r} \times (Id\vec{l} \times \vec{B}) \quad (37)$$

$$= IB\hat{z} \int_0^a r \underbrace{dr}_{=dl} = IB\hat{z} \left[\frac{r^2}{2} \right]_0^a = IB\frac{a^2}{2}\hat{z} = \frac{B^2a^4\omega}{4R}\hat{z}. \quad (38)$$

- e) The power dissipated can be computed simply by considering the mechanical aspect,

$$P = M\omega = \frac{B^2a^4\omega^2}{4R} \quad (39)$$

or by calculating the dissipated Joule heating in the resistor,

$$P = I^2 R = \frac{B^2 a^4 \omega^2}{4R^2} R = \frac{B^2 a^4 \omega^2}{4R}. \quad (40)$$

Exercise 4: Mutual inductance in an AC circuit

a) The self-inductance L is defined as

$$L = \frac{\Phi}{I}, \quad (41)$$

where $\Phi = NBA$ is the magnetic flux through the coil with N windings, with $A = \pi R^2$ being the transverse area of the coil. Knowing the B-field produced in the coil center of $B = \mu_0 \frac{N}{\ell} I$, the self-inductance can be computed:

$$L = \frac{\Phi}{I}, \quad (42)$$

$$= \frac{NBA}{I}, \quad (43)$$

$$= \frac{\mu_0 N^2 \pi R^2 I}{\ell I}, \quad (44)$$

$$= \mu_0 \frac{N^2 \pi R^2}{\ell}, \quad (45)$$

$$(46)$$

b) From above we know that the magnetic field inside a solenoid is given by

$$B = \mu_0 \frac{N}{\ell} I, \quad (47)$$

where ℓ is the length of the coil, while its inductance is equal

$$L = \mu_0 \frac{\pi R^2 N^2}{\ell}, \quad (48)$$

Eliminating ℓ from these two equations we arrive at

$$B = \frac{LI}{\pi R^2 N}. \quad (49)$$

c) The mutual inductance can be calculated starting from its definition

$$M = \frac{\Phi_{21}}{I_1}, \quad (50)$$

where Φ_2 is the magnetic flux produced in the coil 2 by the current I_1 in the coil 1. Since the field lines produced in one coil are all guided into the other coil and the transverse sizes of the

coils are the same we have that the field B_{21} in coil 2 produced by the current I_1 is equal to the field B_{11} in coil 1 produced by I_1 , i.e. $B_{21} = B_{11}$. The flux Φ_{21} thus can be written as

$$\Phi_{21} = \pi R^2 N_2 B_{21} \quad (51)$$

$$= \pi R^2 N_2 B_{11} \quad (52)$$

$$= \pi R^2 N_2 \frac{L_1 I_1}{\pi R^2 N_1} \quad (53)$$

$$= \frac{N_2}{N_1} L_1 I_1. \quad (54)$$

The mutual conductance is thus

$$M = \frac{N_2}{N_1} L_1. \quad (55)$$

Similarly, we can compute M from the flux induced in coil 1 by the current I_2 in coil 2, resulting in

$$M = \frac{N_1}{N_2} L_2. \quad (56)$$

Combining the Eqs. (55) and (56) we obtain

$$M^2 = \frac{N_2}{N_1} L_1 \cdot \frac{N_1}{N_2} L_2 \quad (57)$$

$$= L_1 L_2, \quad (58)$$

$$(59)$$

so that $M = \sqrt{L_1 L_2}$.

- d) By applying the Kirchhoff's second law to the two loops in the circuit we find that the two differential equations for currents I_1 and I_2 are

$$\begin{cases} V_0 \cos(\omega t) - R_1 I_1 - \frac{1}{C} \int I_1 dt - L_1 \frac{dI_1}{dt} - M \frac{dI_2}{dt} = 0, \\ -R_2 I_2 - L_2 \frac{dI_2}{dt} - M \frac{dI_1}{dt} = 0. \end{cases} \quad (60)$$

Exercise 5: Energy in an inductor

- a) From Kirchhoff's rules we find:

$$V - I_1 R_1 - \dot{I}_L L = 0, \quad (61)$$

$$I_2 R_2 - \dot{I}_L L = 0, \quad (62)$$

$$I_1 = I_L + I_2. \quad (63)$$

So we can write:

$$I_2 = \dot{I}_L \frac{L}{R_2}, \quad (64)$$

$$I_1 = I_L + \dot{I}_L \frac{L}{R_2}, \quad (65)$$

and

$$V - \dot{I}_L L - \left[I_L + \dot{I}_L \frac{L}{R_2} \right] R_1 = 0. \quad (66)$$

If we define $L' = L \frac{R_1 + R_2}{R_2}$, then:

$$V - L' \dot{I}_L - I_L R_1 = 0. \quad (67)$$

We can separate variables and integrate, finding:

$$\frac{dI_L}{V - I_L R_1} = \frac{dt}{L'}, \quad (68)$$

$$\Rightarrow \ln(V - I_L R_1) = -\frac{R_1}{L'} t + C, \quad (69)$$

$$\Rightarrow R_1 I_L(t) = V - A e^{-\frac{t}{\tau}}, \quad (70)$$

with $\tau = \frac{L'}{R_1} = L \frac{R_1 + R_2}{R_1 R_2}$ and $A = e^C$, with C the integration constant. We can find the integration constant by looking at the solution at $t = 0$:

$$R_1 I_L(0) = V - A. \quad (71)$$

At $t = 0$ (when the switch is closed), the current suddenly changes from 0, so due to the $\dot{I}_L$ term (i.e. because of the self induction, which is against this change of the current) the whole current passes through R_2 , and it follows $I_L(0) = 0$. Therefore we find:

$$A = V \Rightarrow I_L(t) = \frac{V}{R_1} \left(1 - e^{-\frac{t}{\tau}} \right). \quad (72)$$

Note that for $t \rightarrow \infty$, the circuit is in a steady state in which the whole current flows through the (resistanceless) inductor: $I_L(\infty) = I_1(\infty) = V/R_1$. We can find I_2 using equation (64):

$$I_2(t) = \frac{V L}{R_1 R_2} \left(\frac{1}{\tau} \right) e^{-\frac{t}{\tau}}. \quad (73)$$

Substituting τ :

$$I_2(t) = \frac{V L}{L' R_2} e^{-\frac{t}{\tau}} \quad (74)$$

$$= \frac{V}{R_1 + R_2} e^{-\frac{t}{\tau}}. \quad (75)$$

So, immediately after closing the switch we have

$$I_2(0) = \frac{V}{R_1 + R_2} = 0.91 \text{ A}. \quad (76)$$

And after it remains closed for a long time we have

$$I_2(\infty) = 0 \text{ A}. \quad (77)$$

We integrate now the power at resistance R_2 to obtain the dissipated heat:

$$W_{R_2} = \int_0^\infty I_2^2(t) R_2 dt = \frac{R_2 V^2}{(R_1 + R_2)^2} \int_0^\infty e^{-\frac{2t}{\tau}} dt = \frac{L V^2}{2 R_1 (R_1 + R_2)} = 45.5 \text{ J}. \quad (78)$$

b) As the switch is opened again, the current through the inductor is:

$$I_L(\infty) = \frac{V}{R_1} = 10 \text{ A}. \quad (79)$$

The entire energy stored in the inductor will be dissipated as heat in resistor R_2 . Hence it is:

$$W'_{R_2} = \frac{1}{2} L I_L^2(\infty) = 500 \text{ J}. \quad (80)$$